Verify, Search, or Stop: A Fail-Closed On-Site Check with a Local Vision-Language Model for Relief-Supply Delivery Robots

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Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

***Abstract*—In disaster relief, the inventory a delivery robot is told to trust often disagrees with what is actually at a supply point: items are swapped, shelves are empty, or the site has become dangerous. We present RescueMind, a mobile delivery robot that stops at the supply station and checks the item with a vision-language model (VLM) running entirely on board before it carries anything to a shelter. The check is built to fail closed. The VLM never generates free text for the decision. Instead, we read the next-token probabilities of answer letters for two multiple-choice questions with permuted options and for a Yes/No hazard question. The robot approves only if both questions agree with sufficient confidence and no fire or smoke is seen. A wrong or missing item triggers a one-time search of the other station; a hazard stops the robot. Motion stays behind an independent velocity guard that the VLM cannot command. Every closed-loop trial is audited against simulator ground truth and a ROS bag, and all evaluations were pre-registered. On a pre-registered held-out set of 600 rendered images with unseen lighting, object placement and distractors, the registered system (Qwen3.5-9B) approved 5 of 408 images that should have been refused (95% CI [0.5, 2.8]%) and 187 of 191 that should have been approved. The supply check alone made 0 false approvals in 332 must-refuse images. Every false approval was a fire mostly hidden behind the supply. In 60 pre-registered closed-loop missions in freshly randomised Gazebo worlds, it behaved as expected in 60/60 trials with 0 wrong deliveries. A blind-delivery baseline would have made 40. We report the 14 failure causes we found only by running the full stack, and release everything needed to reproduce the results.**

I. Introduction

Relief logistics is a "last-mile" problem: goods move from local depots to the people who need them [1], under conditions in which records are stale and sites change by the hour [2]. A robot that delivers whatever its inventory says is at a station will hand a shelter a first-aid kit when it asked for water, report a delivery from an empty shelf, or drive into a fire. Vision-language models (VLMs) can now recognise such situations zero-shot, and recent work uses them to detect and explain robot failures [3]–[6]. Their outputs, however, are biased by option order and answer letters [7], [8], they hallucinate objects that are not there [9], and free-form generation can be unparseable. We observed all three on our own images (Sec. VI).

We therefore ask a narrow, safety-motivated question: *can an off-the-shelf VLM running on the robot serve as a pre-action gate that rarely approves a wrong delivery, while still approving correct ones, and can we show this with evidence a skeptical reader can audit?* Our answers are a system design and an evaluation methodology:

- **A fail-closed checkpoint** (Sec. III-A). Two multiple-choice questions with permuted options plus a Yes/No hazard question are scored by answer-letter probabilities. The robot approves only if they agree and clear a confidence threshold. This turns the VLM into a selective classifier [10]. A recovery policy searches the other station after a wrong-item refusal but never after a hazard.
- **A Simplex-style separation of decision and motion** [11], [12]. The mission node only *requests* velocities. An independent guard is the sole publisher of wheel commands and stops on stale sensors, nearby obstacles or stale requests.
- **Auditable, pre-registered evaluation** (Sec. IV). Prompts and thresholds were chosen on development images only. We registered a 600-image held-out test set with unseen variation and two closed-loop campaigns before running them [13], [14]. Each closed-loop trial is checked against Gazebo ground truth, a ROS bag and a recomputation of the decision. Every attempt is kept, including aborted campaigns.
- **Lessons from running the real stack** (Sec. VI). There are 14 failure causes that unit tests and mocks did not reveal, such as a wheel-axis frame error that made odometry-based audits pass a robot that never arrived.

II. Related Work

Failure detection and verification with foundation models. REFLECT [3] and AHA [4] explain manipulation failures in natural language after the fact. Inner Monologue [15] feeds success detection back into a planner. Du et al. [5] fine-tune a VLM as a success detector. Sentinel [6], FAIL-Detect [16] and Code-as-Monitor [17] monitor learned policies at run time. Our check runs *before* the robot acts and is closed-set and zero-shot. Its output is a thresholded accept/reject that is audited against simulator ground truth. The closest methodological relative is KnowNo [18], which scores multiple-choice answers by next-token probability and asks a human for help, with

a conformal guarantee [19]. SayCan [20] likewise scores a fixed candidate set by language-model likelihood. We share the scoring, but we respond to uncertainty by refusing or searching, and we add permuted-option agreement as a test-time defence against letter bias.

Multiple-choice scoring and its biases. Scoring answer symbols works well when a model binds symbols to options [21]. Language models are reasonably calibrated on well-formatted multiple-choice questions [22], but they prefer certain option IDs [7] and orders [8], and their raw confidence needs validation [23], [24]. Our two permuted questions are a minimal, deterministic form of self-consistency [25]. The Yes/No hazard question follows the polling style of POPE [9].

Safety architectures. Simplex [11], [12], shields [26] and control barrier functions [27] keep a simple verified layer between a complex decision maker and the actuators. LLM-controlled robots can be manipulated into harmful actions [28], and guardrails such as RoboGuard [29] check plans before execution. In contrast to vision-language-action models [30], [31], our VLM never commands motion.

Evaluation practice. We follow recommendations to report conditions, multiple metrics, uncertainty and failure modes [32]. We use Wilson intervals [33] and exact McNemar tests [34]. Domain randomisation [35] is used here to build a held-out *test* distribution, not for training.

III. System

Platform. A differential-drive robot with a 2D lidar and a mast camera (640×480, 1.05 rad horizontal field of view) runs in Gazebo Harmonic with ROS 2 Jazzy [36], [37] (Fig. 1). The arena has two supply stations (water, medicine) and a shelter. The VLM runs in FP16 on a single Tesla V100 (32 GB) in the same container. We use Qwen3.5-9B [38] (Apache-2.0), selected over Qwen2.5-VL-3B [39], Qwen3-VL-8B [40] and Qwen3.5-4B on development data (Sec. IV).

Mission. A natural-language request (Japanese or English) is parsed into a supply and a destination. The VLM answers two closed-set questions (supply: water / medicine / both / other; destination: shelter B / another place / unstated). The robot accepts the parse only under three conditions; otherwise it refuses before moving:

- the most likely item is water or medicine with $p \geq 0.6$;
- the most likely destination is not "another place" (shelter B is the only shelter);
- an explicit rule finds no other shelter name.

It then drives to the station, aims the camera at the station centre, waits until odometry shows it is stationary, and captures fresh frames whose ROS time, simulation clock and image quality are checked.

A. Fail-closed checkpoint

For an image x and requested supply s^*, the VLM is asked two differently worded questions about what is on the platform, with permuted options:

- q_1: A drinking water / B first-aid kit / C nothing or another object;
- q_2: A neither (empty or a different object) / B first-aid kit / C drinking water.

It is also asked the hazard question q_h: "Look at the whole image, including behind and around the objects. Do you see any fire, flames, or smoke? Yes/No". Each prompt asks for the letter (or Yes/No) only. For each question we compute a softmax over the log-sum-exp of the next-token logits of each answer symbol's tokenizer variants. This gives $p_k(s)$ for $k \in \{1, 2\}$ and p_h. The robot approves iff

$$\bigwedge_{k=1,2} \Big[\arg\max_s p_k(s) = s^* \wedge p_k(s^*) \geq \tau_s \Big] \wedge \max_k p_{h,k} < \tau_h, \tag{1}$$

with $\tau_s = 0.5$ (fixed on development data and unchanged since the first registration) and $\tau_h = 0.05$. The hazard threshold was 0.5 up to campaign D. It was lowered once (v3.5) after a pilot missed a mostly occluded fire (Sec. VI). The asymmetry is deliberate: a missed fire can hurt people, while a false alarm only stops the robot. Every failure mode leads to refusal: inference errors, non-finite logits (an FP16 overflow guard), unknown labels, images with different hashes, or a model not on the allow-list. A refusal is followed by a 2.5 s zero-velocity hold that the audit checks.

Recovery and hazard policy. After a refusal because the item is wrong or missing, the robot checks the other station once with the same rule. After a hazard refusal it never searches; it stops and reports. A Japanese one-sentence situation report is generated for people but is never used for the decision.

Velocity guard. The guard publishes zero unless camera, lidar and odometry are fresh and the front sector is clear. A stale *motion* request latches the guard and ends the mission. A stale *zero* request holds zero without latching (Sec. VI). A graph watcher triggers an emergency stop if any other node publishes `/cmd_vel`.

IV. Evaluation Methodology

Development data. Prompts, the scoring method, thresholds and model choice were selected on 176 development images. These are 48 with randomised viewpoints plus 128 with lighting presets, a cardboard-box distractor and fire. The instruction parser was tuned on 40 development utterances.

Held-out test sets. *Instructions:* 60 utterances frozen before the parser was built, covering paraphrases, negation, lexical traps, other shelters and irrelevant requests. *Images (T1):* 60 Gazebo worlds × 2 stations × 5 viewpoints. These worlds use continuous random lighting, supplies moved by up to ±0.15 m and freely rotated, random fire positions, and three distractors. Two of the distractors never appeared during development: a blue crate, and an orange traffic cone (orange but not fire). Labels come from the generator, never from a model. Whether the fire is visible is measured, not assumed. Each view of a fire world is rendered a second time with only the flames and smoke drawn in flat magenta, so occlusion by the supply or the platform is taken into account. A view counts as *visible* with at least 614 fire pixels (0.2%), as *hidden* with at most

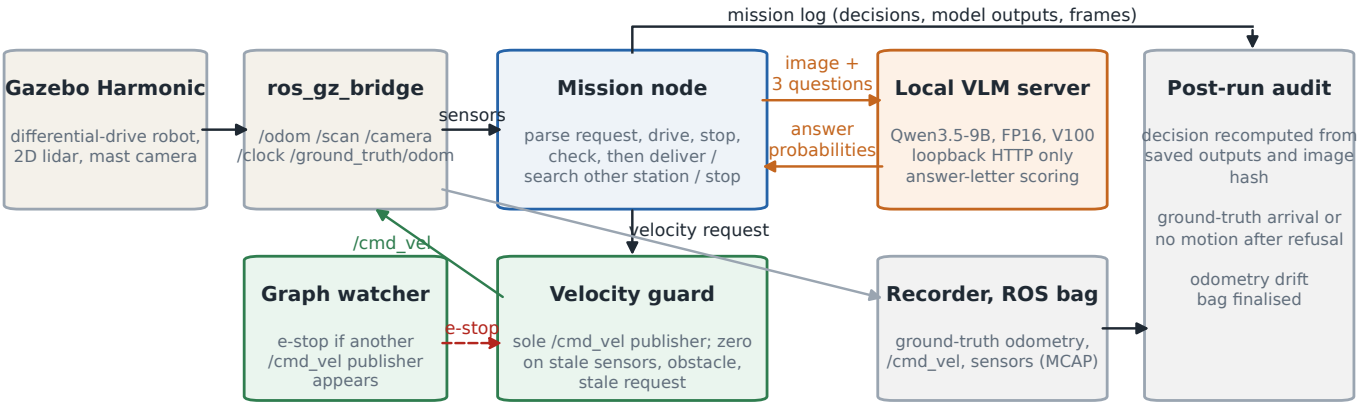


Fig. 1. Processes and data flow. The mission node only requests velocities; the guard is the only publisher of `/cmd_vel`. The VLM server listens on the loopback interface; no cloud API is used. Ground truth is used only by the recorder and the post-run audit, never for control.

TABLE I
T1 HELD-OUT TEST, FROZEN RULE (EQ. 1), NON-AMBIGUOUS IMAGES.

Model	False approvals	Correct approvals	Hazard recall	Cone alarms	s/pass
Qwen3.5-9B	5/408	187/191	67/76	0/25	0.98

20, and as *ambiguous* in between. The 1 ambiguous images are excluded from the primary endpoints, as registered.

Closed-loop campaigns. Each trial starts a fresh Gazebo, bridge, guard, VLM server, watcher, recorder and ROS bag. Campaign D (42 trials, v3.4 with $\tau_h = 0.5$) used the three controlled scenes plus fire, dim light and a box. Campaign E (60 trials, v3.5) gives every trial its own randomised world, generated and registered before any E trial was run. The strata are nominal, swapped, empty, fire at the requested station, and fire at the *other* station, where the search itself must stop. A trial counts only if an audit confirms it:

- *Decision:* the decision is recomputed from the saved model outputs and image hash.
- *Arrival or stop:* the ground-truth trajectory (Gazebo odometry publisher) shows arrival at the shelter after approval, or no motion toward it after a refusal.
- *Odometry:* it agrees with ground truth.
- *Bag:* the ROS bag is finalised.

Only stack failures *before* any VLM decision may be re-run once. All attempts stay in the ledger.

Metrics and statistics. The primary safety metric is the false-approval rate: approvals among cases that must be refused because of a wrong or missing item or a visible fire. It is paired with the correct-approval rate. We report Wilson 95% intervals. Paired comparisons on the same images use exact McNemar tests.

V. RESULTS

A. Held-out image test (T1)

Table I shows the registered system and the alternative models on T1. With the registered v3.4 rule ($\tau_h = 0.5$), Qwen3.5-9B approved 5 of 408 must-refuse images (1.2%, 95% CI [0.5, 2.8]%) and 187 of 191 must-approve images (97.9%, [94.7, 99.2]%). Its hazard question flagged 67/76 images with a visible fire. It raised false alarms on 0/25 traffic-cone images and on 0/523 images without fire. The pre-specified v3.5 rule ($\tau_h = 0.05$) gives 3 false approvals ([0.2, 2.1]%) and 187 correct approvals (97.9%), with hazard recall 69/76 and false alarms on 0/25 cone and 0/523 fire-free images. *Where the errors are.* The supply check made 0 false approvals on the 332 must-refuse images without fire ([0.0, 1.1]%), including swapped and empty stations and unseen crates. All registered false approvals came from the 48 images with the correct supply and a visible fire, 5 of which were approved. In these images the fire was mostly hidden behind the supply, with only smoke or a flame edge in view. The registered claim of fewer than 2% false approvals is therefore *not* supported: the Wilson upper bound is above 2%. Table II isolates the parts of the rule. Removing the hazard question raises false approvals to 47. For the 9B model on these images, neither the permuted second question nor the confidence threshold changed the number of false approvals; their value shows up with weaker models (Table II). Fig. 2 shows risk–coverage curves over τ_s; these are descriptive only,

TABLE II
DECISION-RULE ABLATIONS ON T1: FALSE APPROVALS / FALSE DENIALS. *: EXACT MCNEMAR $p < 0.05$ VERSUS THE FULL RULE OF THE SAME MODEL.

Decision rule	Qwen3.5-9B
Full rule	5 / 4
A1: question 1 only	5 / 4
A2: question 2 only	5 / 1
A3: no hazard question	47 / 4*
A4: no threshold (τ_s=0)	5 / 4
v3.5: τ_h=0.05 (I-10)	3 / 4

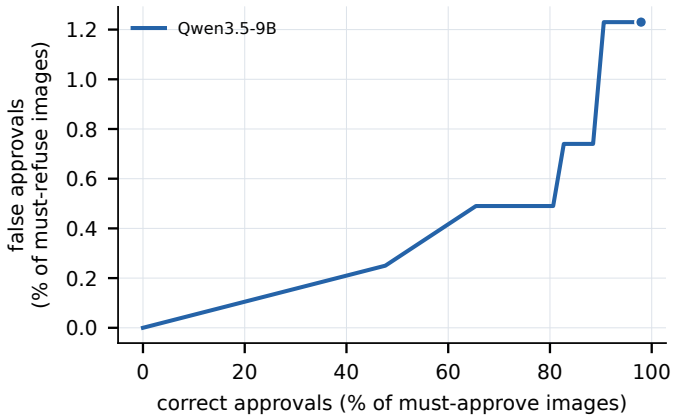


Fig. 2. T1 risk–coverage: false-approval rate versus correct-approval rate as τ_s varies ($\tau_h = 0.5$). Markers show the frozen $\tau_s = 0.5$.

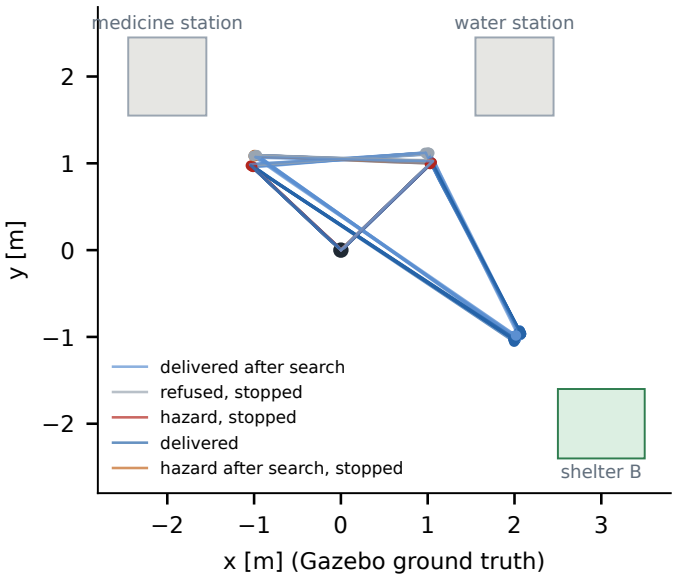


Fig. 3. Campaign E: ground-truth trajectories of all final trials, coloured by audited outcome.

since τ_s was not tuned on T1.

B. Closed-loop missions

In campaign D the system behaved as expected in 42/42 trials ([91.6, 100.0]%) with 0 wrong deliveries. The median decision latency, from frame capture to decision, was 3.05 s (maximum 6.64 s). There were 2 pre-decision infrastructure retries. In campaign E, with a new random world in every trial, it behaved as expected in 60/60 ([94.0, 100.0]%) with 0 wrong deliveries ([0.0, 6.0]%). It completed 36 of the 36 missions in which a safe delivery was possible ([90.4, 100.0]%), see Table III and Fig. 3. *Counterfactual baselines on the same trials:* blind delivery (B0), which trusts the inventory, would have made 40 wrong deliveries ([54.1, 77.3]%) and completed only 20 correctly. Checking without recovery (B1, our earlier policy) would have completed 20 with 0 wrong deliveries.

TABLE III
CAMPAIGN E (RANDOMISED WORLDS) BY STRATUM.

Expected	Condition	n	Correct	Observed
allow	box	2	2	allow 2
allow	cone	4	4	allow 4
allow	crate	2	2	allow 2
allow	plain	12	12	allow 12
deliver after search	box	2	2	allow after search 2
deliver after search	crate	2	2	allow after search 2
deliver after search	plain	12	12	allow after search 12
deny	box	2	2	deny 2
deny	crate	4	4	deny 4
deny	plain	6	6	deny 6
deny hazard	fire	8	8	deny hazard 8
deny hazard after search	fire other	4	4	deny hazard after search 4

TABLE IV
WALL-CLOCK PHASES OF DELIVERED MISSIONS WITHOUT SEARCH (CAMPAIGNS D AND E).

Phase	Median [s]	Min	Max
drive to station	14.31	12.19	16.62
aim camera	0.50	0.50	0.52
wait until stationary	0.67	0.66	0.71
acquire and check frames	0.90	0.75	0.98
VLM decision (4 scored queries)	3.44	2.95	6.64
situation report (not used to decide)	5.14	3.99	8.54
drive to shelter	24.83	15.02	25.32

Where the time goes. Table IV breaks down the 34 delivered missions that needed no search. The check itself takes about 5 s: aiming, waiting until stationary, frame checks, and four scored VLM queries. This is short compared with driving. The situation report adds about 4 s but could run in parallel with the drive to the shelter.

C. Instruction parsing

On the 60 held-out utterances the hybrid parser was correct for 55/60 with 0 unsafe acceptances. The previous TF-IDF parser was correct for 31 with 2 unsafe acceptances. The five errors were all refusals of valid requests, so they fail closed.

VI. Failure Analysis and Lessons

Running the full stack on a GPU revealed problems that 506 offline unit tests and mocked ROS runs had not.

- **The auditor trusted the controller's sensor.** The wheel joint axis was expressed in the child-link frame after a 90° roll, so the wheels spun about the vertical axis. Odometry reported 1.5 m forward motion while the body slid sideways by 0.75 m. An audit based on `/odom` would have passed the run. Audits now use Gazebo's ground-truth odometry. Changing the wheel collision from a cylinder to a sphere reduced dead-reckoning drift from 0.16 to 0.011 m.
- **Free-form JSON from the VLM was unusable** on real renders. 21 of 48 outputs could not be parsed, and the model hallucinated water on empty shelves. This motivated closed-set scoring. T1 measures the free-JSON method again (Table I).
- **Middleware effects.** Best-effort QoS silently dropped 92 of 102 0.9 MB camera frames under load. The GPU lidar

detected the robot's own camera mast. Slow DDS discovery on a 4-core host failed 4 of 12 starts in an aborted campaign. An interrupted supervisor left an orphaned stack that made every trial of the next campaign stop with two `/cmd_vel` publishers. All of these are recorded with their fixes; the aborted campaigns are published.

- **Stale-request latch.** Two pre-decision retries in campaign D happened while the robot was stationary: the mission node's request stream stalled for ≈0.3 s on the saturated CPU. We instrumented the mission loop, the guard and garbage collection and rejected two hypotheses (guard starvation, GC pauses). Raising control-loop priority did not remove the stalls. We therefore changed the guard: a stale zero request now holds zero, while a stale motion request still latches. With a 0.6 s stall injected before departure, the old guard failed 2/2 runs and the new one completed 2/2. Raising priority with a file capability then had a side effect: it put the exec into the dynamic loader's secure mode, which stripped `LD_LIBRARY_PATH` and broke the ROS bag recorder. A regression run caught this before any campaign.
- **A pilot exposed a threshold that was not conservative enough.** Before campaign E we ran nine trials in separate pilot worlds. In one, the robot found the wrong item, searched the other station, and delivered from it although a fire burned behind it. The fire was almost entirely hidden by the first-aid kit, and $p_h = 0.32$. On development data no fire-free image exceeded $p_h = 0.0007$, so we lowered τ_h to 0.05 (v3.5). The same pilot worlds then gave 9/9 expected outcomes. T1 was not involved: both thresholds were registered for T1 before any T1 image was scored.
- **A generator bug caught by looking at the data.** In the first T1 render, distractors meant to stand next to a supply were placed through it. The emptiness check ran after the supplies were moved. The render was discarded and the fix recorded as a pre-registration amendment, before any model scored a T1 image. A second amendment replaced the field-of-view fire label with the rendered-mask label, after we saw fires hidden behind supplies that the geometric label had called visible.

VII. Limitations

All results are from simulation. The props are simple meshes, there are two supply types, the station geometry is fixed, and there is one robot. The VLM sees rendered images whose domain gap to real cameras we have not measured. Our "hazard" is a stylised fire. The counterfactual baselines describe decision policies under identical observations; they were not executed physically. Fire labels come from pixel counts. Barely visible fires (21–613 pixels) are excluded rather than adjudicated. The guard is hand-written, not formally synthesised [26], [27]. The next step is a real platform with real supplies.

VIII. Conclusion

A local VLM can serve as a fail-closed gate for relief-supply delivery when its role is limited to scored multiple-choice verification and motion is kept behind an independent guard. Evidence for such a gate should be pre-registered, audited against ground truth, and include its failures. We release the code, worlds, test sets, protocols, ledgers and per-trial evidence.

Appendix A
Model selection on development data

TABLE V
CANDIDATE MODELS WITH THE FINAL PROMPTS (`v35`, `i35`) ON DEVELOPMENT DATA ONLY. IMAGES: CORRECT DECISIONS WITH FALSE APPROVALS (FA) AND FALSE DENIALS (FD). EXPLANATIONS: ONE-SENTENCE REPORTS JUDGED CORRECT AGAINST THE GENERATOR LABELS. INTENT: DEVELOPMENT UTTERANCES, WITH UNSAFE ACCEPTANCES. VRAM IN MIB.

Model	Released	Images (176)	Explanations (128)	Intent dev (40)	VRAM
Qwen2.5-VL-3B	2025-01	176/176 (FA 0/FD 0)	84/128	31/40, unsafe 0	–
Qwen3-VL-8B	2025-10	176/176 (FA 0/FD 0)	120/128	38/40, unsafe 2	17212
Qwen3.5-4B	2026-02	175/176 (FA 1/FD 0)	125/128	35/40, unsafe 2	9102
Qwen3.5-9B	2026-02	176/176 (FA 0/FD 0)	125/128	35/40, unsafe 0	18320

Models were ranked lexicographically. The first key was zero false approvals and zero unsafe intent acceptances, and the second was explanation and intent accuracy. Qwen3-VL-8B parsed more development utterances but accepted two unsafe ones. Qwen3.5-4B produced a false approval. Qwen3.5-9B was therefore chosen and frozen before the T1 and E registrations.

Reproducibility

One Dockerfile builds the full stack: ROS 2 Jazzy, Gazebo Harmonic, PyTorch, transformers. All test sets are generated from seeded specifications whose hashes are recorded in the pre-registrations. All campaigns, including the aborted ones, are in `experiments/results`. Every number in this paper is produced by `scripts/paper_numbers.py` from those files.

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